

Network Traffic Forensics Investigation

Simulated PCAP-style analysis: suspicious DNS, ICMP, and connection behavior

Case Summary

This simulated investigation examines abnormal network activity observed during a workstation security alert. No real organization, user, or production system is represented. Example addresses use documentation ranges.

Field	Value
Case ID	NF-2026-002
Endpoint	WS-042 (simulated)
Internal IP	10.20.14.42
External example IP	203.0.113.77
Suspicious domain	update-check.example
Severity	SEV-3 / Medium
Primary evidence	Synthetic PCAP summary, DNS logs, firewall logs
Outcome	Suspicious activity contained; host referred for endpoint investigation

Evidence Review

Evidence	Observation	Interpretation	Next action
DNS queries	Repeated lookups to update-check.example at short, regular intervals.	Beacon-like periodicity; requires correlation before labeling malicious.	Correlate endpoint process and DNS history
Outbound TCP	Repeated connections to 203.0.113.77 on uncommon destination port.	Unusual external communication from user workstation.	Block if confirmed unauthorized; review process owner
ICMP	Short burst of sequential ICMP traffic across internal addresses.	Could indicate discovery, monitoring, or benign tooling.	Correlate source process and approved scanner inventory
Firewall logs	Outbound session attempts continued after DNS resolution.	Supports active host communication, not passive lookup only.	Escalate for host triage
Post-isolation	Connections stopped after endpoint network containment.	Supports endpoint as source of traffic.	Preserve endpoint evidence

IOC / Evidence Table

Indicator	Type	Source	Status	Analyst note
203.0.113.77	Example IPv4	Firewall / PCAP	Synthetic	Documentation-range address used for portfolio demonstration.
update-check.example	Example domain	DNS log	Synthetic	Reserved example domain; used to demonstrate IOC handling.
WS-042	Asset identifier	EDR / CMDB	Synthetic	Simulated workstation identifier.
10.20.14.42	Private IP	DHCP / firewall	Synthetic	Simulated internal host address.

MITRE ATT&CK; Mapping

Observed behavior	ATT&CK; mapping	Why it fits
Sequential host probing	Tactic: Discovery	Behavior could support environment discovery; further evidence is required before choosing a specific technique.
Periodic external communication	Command-and-control concept	Regular outbound connections may indicate beaconing, but periodicity alone is insufficient to confirm C2.
Account enumeration found during host follow-up	T1087 - Account Discovery	ATT&CK; defines T1087 as attempts to list valid accounts in an environment.
PowerShell execution found during host follow-up	T1059.001 - PowerShell	ATT&CK; maps abuse of PowerShell to the Command and Scripting Interpreter technique family.

Analyst Conclusion

- Network evidence justified endpoint isolation and deeper host review, but did not independently prove malware or attribution.
- The strongest conclusion was that WS-042 generated unusual, unauthorized-looking traffic requiring host-level investigation.
- Evidence was preserved through timestamps, source references, and case identifiers to support repeatability.